

Generating a Fixed-Frequency Sinusoidal Signal from Digital Output

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For some testing purposes we may need a fixed-frequency sinusoidal signal of 440Hz, 1kHz, etc. But most low-cost micro-controllers (MCUs) do not have internal digital-to-analogue converters (DACs). Adding external DACs could be expensive and/or complicated, and these are not always needed. Fortunately, most MCUs can easily produce fixed-frequency square-wave signals with frequencies required for the sinusoidal signals.

Circuit and working

The circuit can be configured in two ways; one that produces a sinusoidal signal with RC filters and another that produces a sinusoidal signal

with a combination of RC and LC filters.

Fig. 1 shows the circuit for producing a sinusoidal-like signal with RC filters. Output pin 33 (RB0) of PIC16F877A (IC1) produces a square-wave signal, for example, with a frequency of 1kHz. This signal is filtered using three RC filters built correspondingly with R2 and C4, R3 and C5, and R4 and C6. The cut-off frequency of these filters is practically equal to the frequency of the square-wave signal produced on pin 33 (RB0). This is also the frequency of the produced sinusoidal signal.

This frequency is calculated using the relationship:

$$F(-3dB) = 1 / (2 \times \pi \times R \times C) = 1 / (6.28 \times R \times C) [Hz]$$

where R is R2, R3 or R4, and C is C4,

C5 or C6.

Also, for good filtration of harmonics, the following equation is suggested:

$$R4 = 10 \times R3 = 100 \times R2$$

For a frequency of 1kHz, the following popular values are suggested:

$$R2 = 1k\text{-ohm}, R3 = 10k\text{-ohm and } R4 = 100k\text{-ohm}$$

$$C4 = 150nF, C5 = 15nF \text{ and } C6 = 1.5nF$$

The filtered signal at point A is amplified by operational amplifier TL071 (IC2) and is available on connector CON2. Gain of the output stage = $1 + R7/R6 = 10$, and is selected according to the requirement.

Fig. 2 shows the circuit for pro-

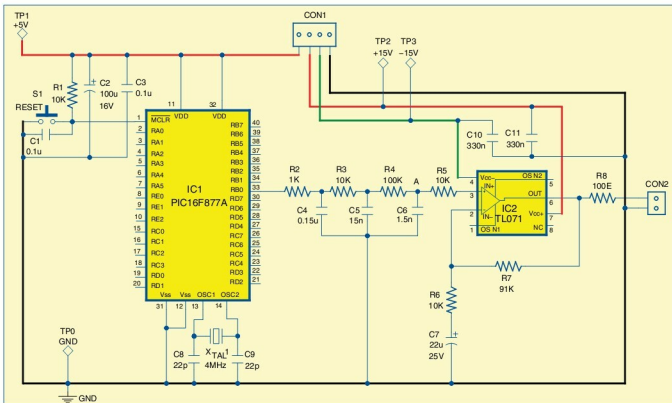


Fig. 1: Circuit diagram for producing a sinusoidal signal with RC filters